

Evaluación Final - Proyecto Heart Disease Dataset

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Base de datos: <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

1. Análisis descriptivo 1 variable cualitativa

Descriptives

	sex
N	920
Perdidos	0
Media	
Mediana	
Desviación estándar	
Mínimo	
Máximo	

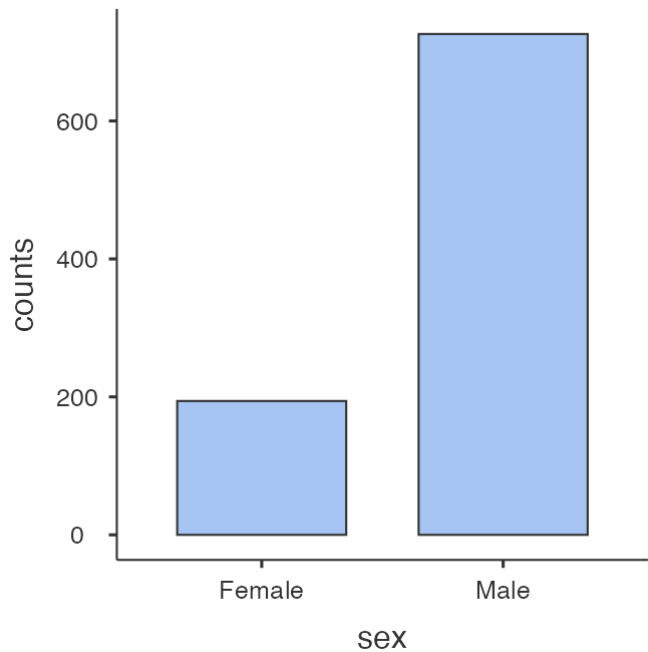
Frequencies

Frequencies of sex

sex	Counts	% of Total	Cumulative %
Female	194	21.1%	21.1%
Male	726	78.9%	100.0%

Plots

sex



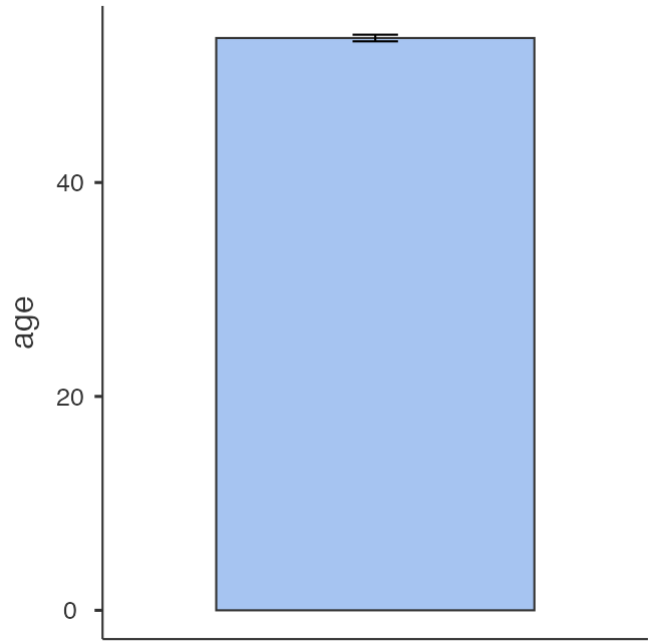
2. Análisis descriptivo 1 variable cuantitativa

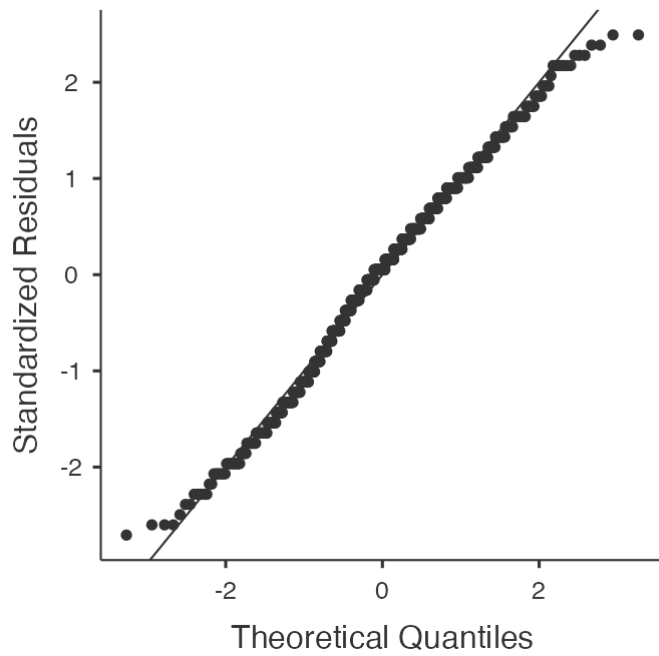
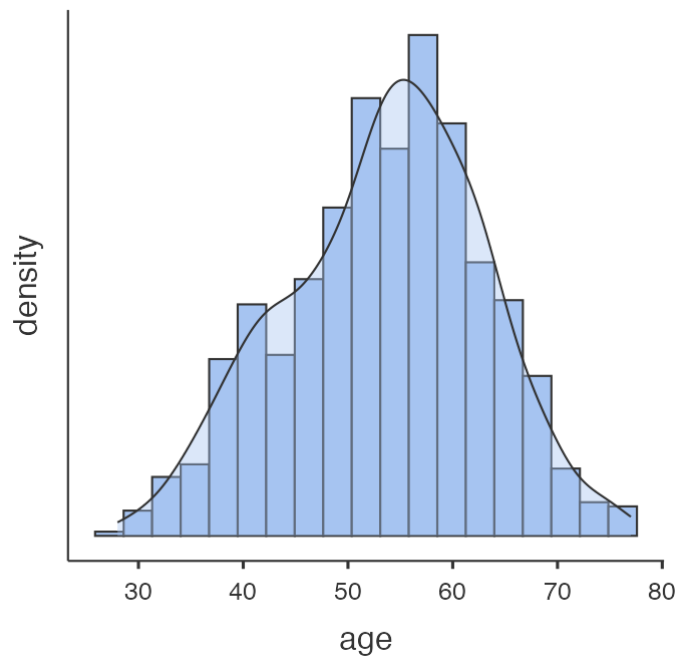
Descriptives

	age
N	920
Perdidos	0
Media	53.5
Mediana	54.0
Moda	54.0
Desviación estándar	9.42
Varianza	88.8
Mínimo	28
Máximo	77

Plots

age





3. Análisis descriptivo 2 variables cualitativas

sex		cp				Total
		asymptomatic	atypical angina	non-anginal	typical angina	
Female	Observado	70	61	53	10	194
	Esperado	105	36.7	43.0	9.70	194
	% de fila	36.1%	31.4%	27.3%	5.2%	100.0%
	% de columna	14.1%	35.1%	26.0%	21.7%	21.1%
Male	Observado	426	113	151	36	726
	Esperado	391	137.3	161.0	36.30	726
	% de fila	58.7%	15.6%	20.8%	5.0%	100.0%
	% de columna	85.9%	64.9%	74.0%	78.3%	78.9%
Total	Observado	496	174	204	46	920
	Esperado	496	174	204	46	920
	% de fila	53.9%	18.9%	22.2%	5.0%	100.0%
	% de columna	100.0%	100.0%	100.0%	100.0%	100.0%

χ² Tests

	Value	df	p
χ²	37.9	3	<.001
N	920		

gender vs test preparation course

4. Análisis descriptivo 2 variable cuantitativas

Correlation Matrix

		age	thalch
age	r de Pearson	—	
	gl	—	
	valor p	—	
thalch	r de Pearson	-0.366***	—
	gl	863	—
	valor p	<.001	—

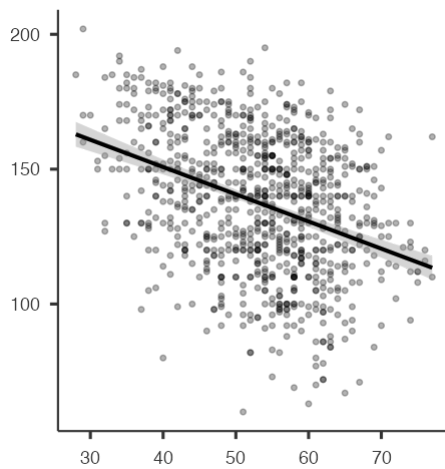
Note. * p < .05, ** p < .01, *** p < .001

Plot

age

thalch

age



thalch

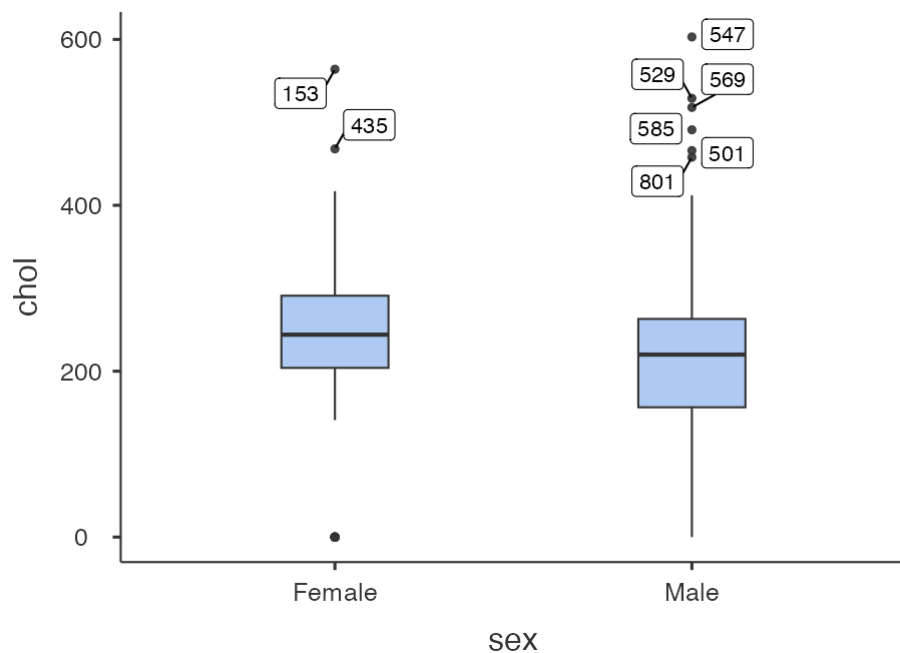
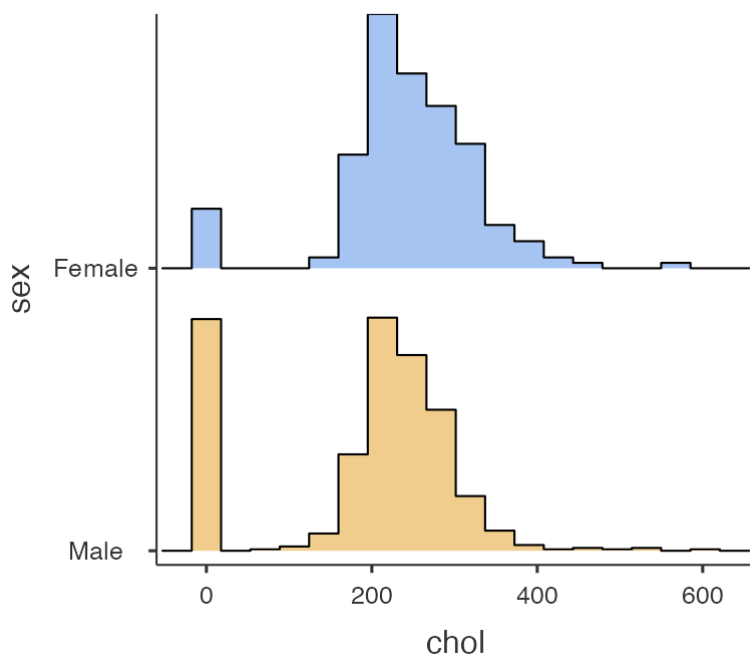
5. Análisis descriptivo 1 variable cuantitativa vs 1 variable cualitativa

Descriptives

	sex	N	Missing	Mean	Median	SD	Minimum	Maximum
chol	Female	187	7	241	244	86.3	0	564
	Male	703	23	188	220	113.9	0	603

Plots

chol



6. Pruebas de Hipótesis sobre medias (pruebas t y prueba ANOVA)

Independent Samples T-Test

						95% Confidence Interval		
		Statistic	df	p	Mean difference	SE difference	Lower	Upper
thalch	T de Student	5.35 ^a	863	<.001	11.2	2.09	7.08	15.3

Note. $H_a: \mu_{\text{Female}} \neq \mu_{\text{Male}}$

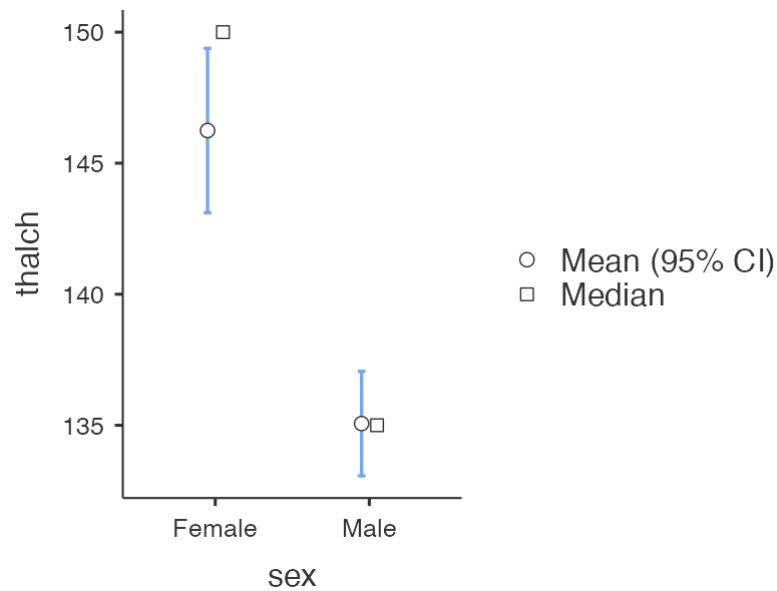
^a Levene's test is significant ($p < .05$), suggesting a violation of the assumption of equal variances

Group Descriptives

	Group	N	Mean	Median	SD	SE
thalch	Female	192	146	150	22.2	1.60
	Male	673	135	135	26.4	1.02

Plots

thalch



T-TEST UNA MUESTRA

One Sample T-Test

		Statistic	df	p	Mean difference
age	T de Student	172	919	<.001	53.5

Note. $H_a: \mu \neq 0$

Descriptives

	N	Mean	Median	SD	SE
age	920	53.5	54.0	9.42	0.311

ANOVA UN FACTOR

One-Way ANOVA (Welch's)

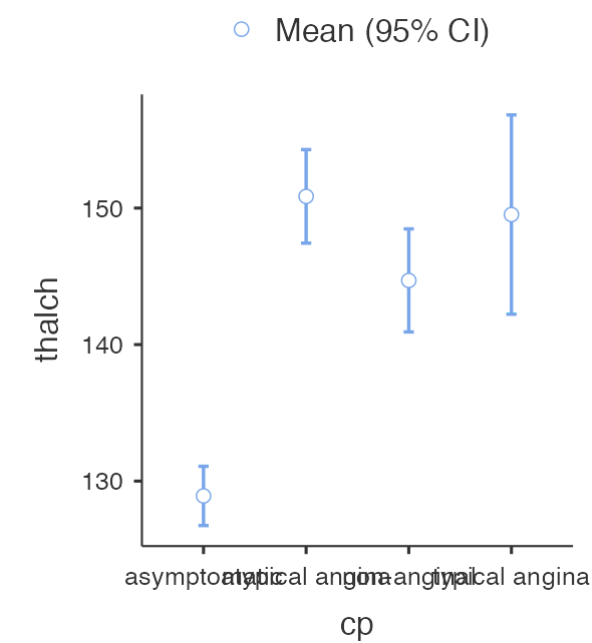
	F	df1	df2	p
thalch	47.9	3	168	<.001

Group Descriptives

cp		N	Mean	SD	SE
thalch	asymptomatic	470	129	23.9	1.10
	atypical angina	167	151	22.4	1.74
	non-anginal	186	145	26.1	1.91
	typical angina	42	150	23.4	3.61

Plots

thalch



Post Hoc Tests

Tukey Post-Hoc Test – thalch

		asymptomatic	atypical angina	non-anginal	typical angina
asymptomatic	Mean difference	—	-21.9	-15.78	-20.61
	p-value	—	<.001	<.001	<.001
atypical angina	Mean difference	—	—	6.16	1.33
	p-value	—	—	0.079	0.989
non-anginal	Mean difference	—	—	—	-4.82
	p-value	—	—	—	0.645
typical angina	Mean difference	—	—	—	—
	p-value	—	—	—	—

ANÁLISIS 7: PCA (ANÁLISIS DE COMPONENTES PRINCIPALES)

Component Loadings

	Component	
	1	Uniqueness
age	0.745	0.445
trestbps	0.439	0.807
chol		0.984
thalch	-0.667	0.555
oldpeak	0.624	0.611
ca	0.657	0.568

Note. 'varimax' rotation was used
[4]

Assumption Checks

Bartlett's Test of Sphericity

χ^2	df	p
203	15	<.001

Iteración 2

Cargas de los Componentes

	Componente	
	1	Unicidad
age	0.741	0.451
trestbps	0.429	0.816
thalch	-0.682	0.535
oldpeak	0.624	0.611
ca	0.658	0.568

Note. Se utilizó la rotación 'varimax'
[4]

Comprobaciones de Supuestos

Prueba de Esfericidad de Bartlett

χ^2	gl	p
190	10	<.001

Iteración 3

Cargas de los Componentes

	Componente	
	1	Unicidad
age	0.720	0.482
thalch	-0.731	0.466
oldpeak	0.624	0.610
ca	0.691	0.522

Note. Se utilizó la rotación ‘varimax’
[4]

Comprobaciones de Supuestos

Prueba de Esfericidad de Bartlett

χ^2	gl	p
152	6	<.001

Iteración 4

Cargas de los Componentes

	Componente	
	1	Unicidad
age	0.802	0.356
thalch	-0.732	0.464
ca	0.713	0.492

Note. Se utilizó la rotación ‘varimax’
[4]

Estadísticas de los Componentes

Resumen

Componente	SC Cargas	% de la Varianza	% Acumulado
1	1.69	56.3	56.3

Correlaciones Entre Componentes

1	
1	—

Comprobaciones de Supuestos

Prueba de Esfericidad de Bartlett

χ^2	gl	p
103	3	<.001

Valores Propios

Valores Propios Iniciales

Componente	Valor Propio	% de la Varianza	% Acumulado
1	1.688	56.3	56.3
2	0.737	24.6	80.8
3	0.575	19.2	100.0

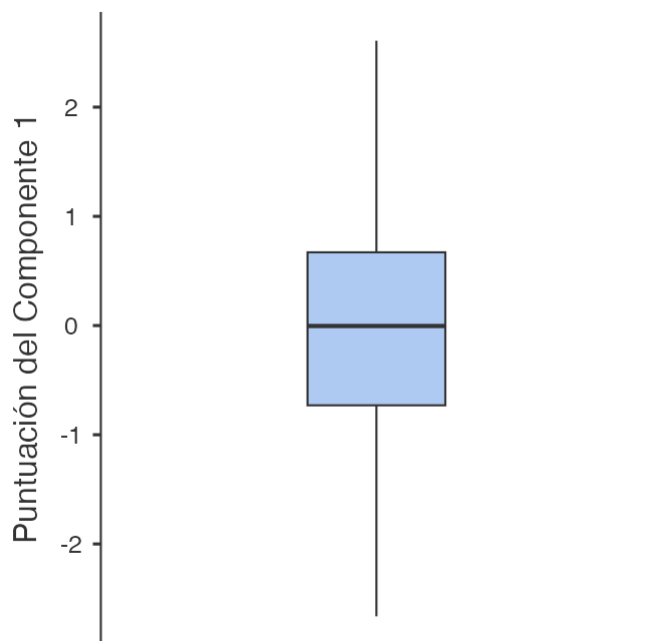
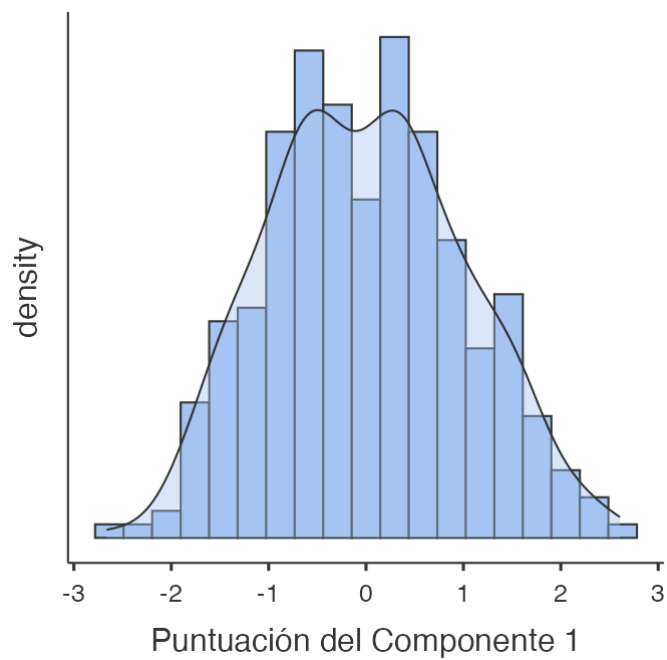
Descriptivas

Descriptivas

Puntuación del Componente 1	
N	309
Perdidos	611
Media	3.42e-16
Mediana	-0.00407
Desviación estándar	1.000
Mínimo	-2.66
Máximo	2.61

Gráficos

Puntuación del Componente 1



8. Regresión Lineal MULTIPLE

Model Fit Measures

Model	R	R ²	Adjusted R ²
1	0.419	0.176	0.172

Note. Models estimated using sample size of N=827

Predictor	Estimate	SE	95% Confidence Interval		t	p
			Lower	Upper		
Constante	183.6920	6.85464	170.2373	197.1466	26.80	<.001
age	-0.8739	0.09353	-1.0575	-0.6903	-9.34	<.001
trestbps	-0.0567	0.04507	-0.1452	0.0318	-1.26	0.209
chol	0.0475	0.00753	0.0327	0.0623	6.30	<.001
oldpeak	-1.7241	0.77902	-3.2532	-0.1950	-2.21	0.027

Assumption Checks

Normality Tests

	Statistic	p
Shapiro-Wilk	0.988	<.001
Kolmogorov-Smirnov	0.0453	0.067
Anderson-Darling	3.09	<.001

Note. Additional results provided by moretests

Heteroskedasticity Tests

	Statistic	p
Breusch-Pagan	4.92	0.296
Goldfeld-Quandt	1.11	0.151
Harrison-McCabe	0.468	0.098

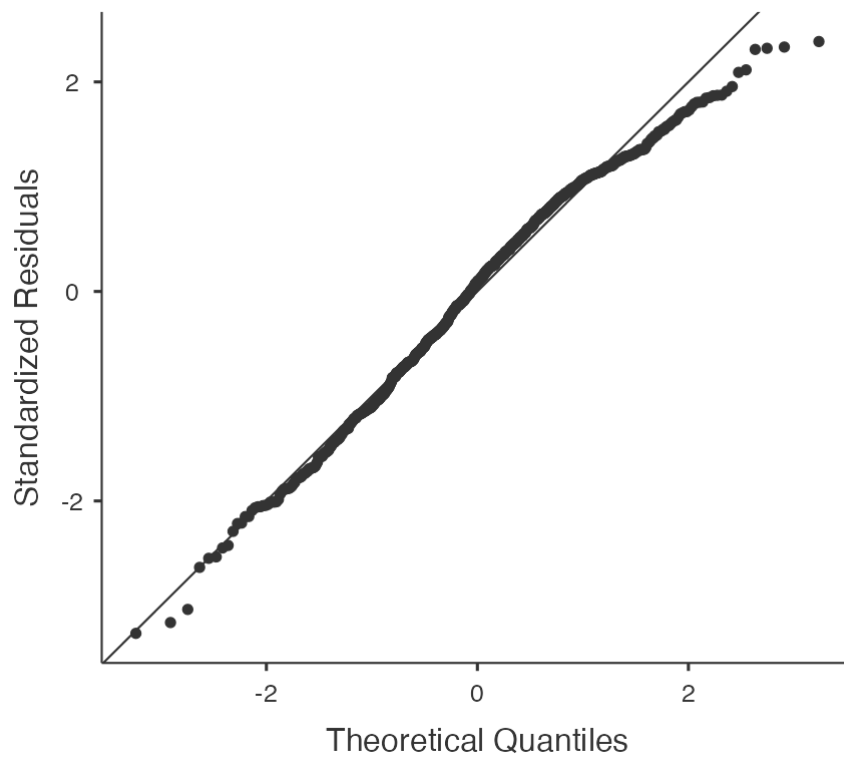
Note. Additional results provided by moretests

Durbin–Watson Test for Autocorrelation

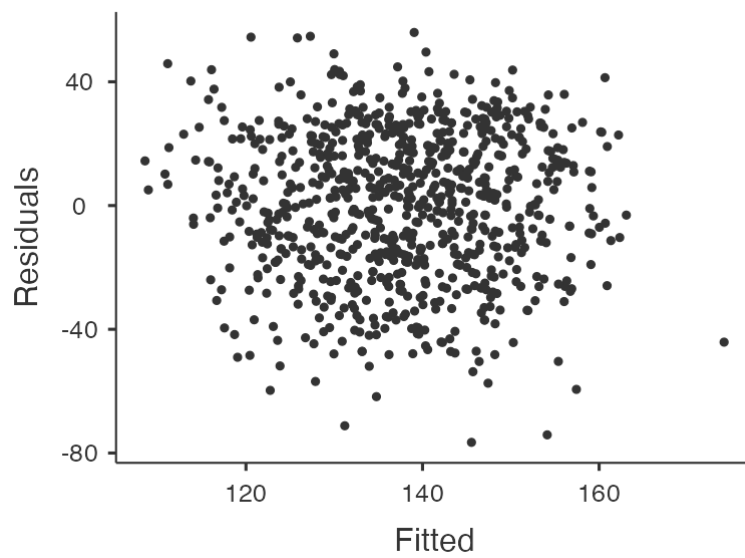
Autocorrelation	DW Statistic	p
0.169	1.66	<.001

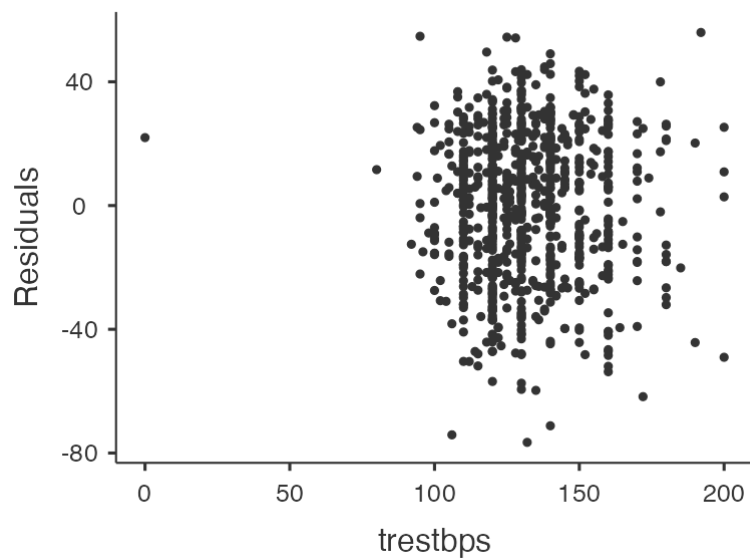
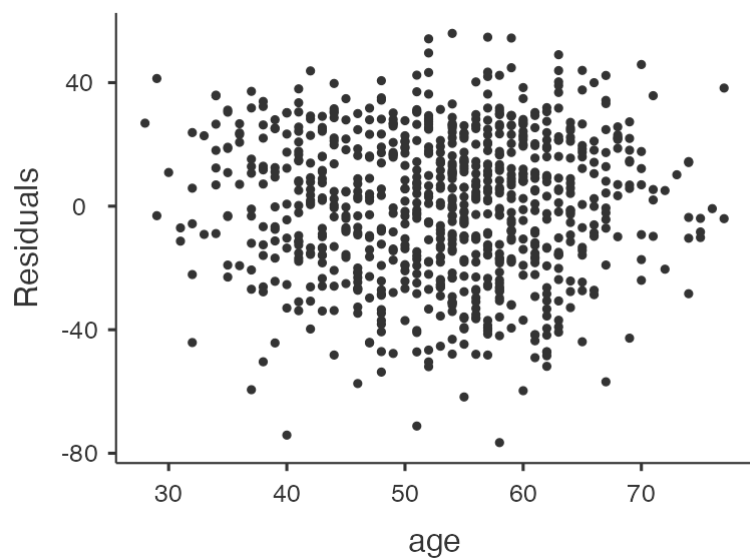
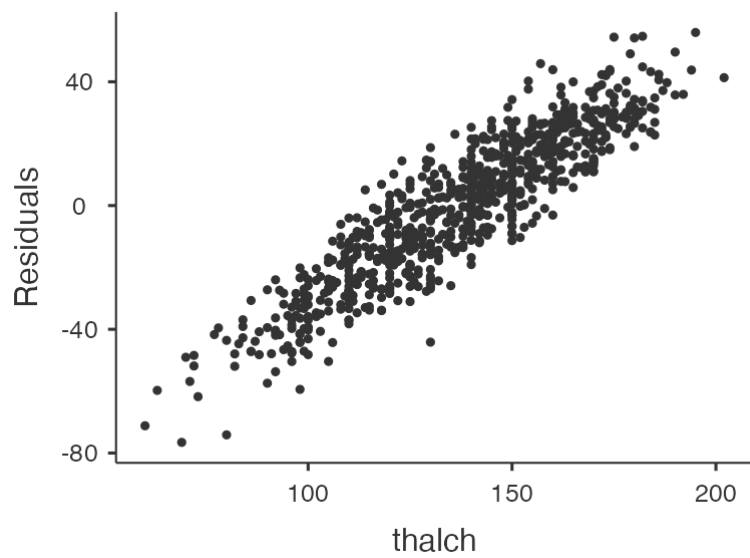
[5]

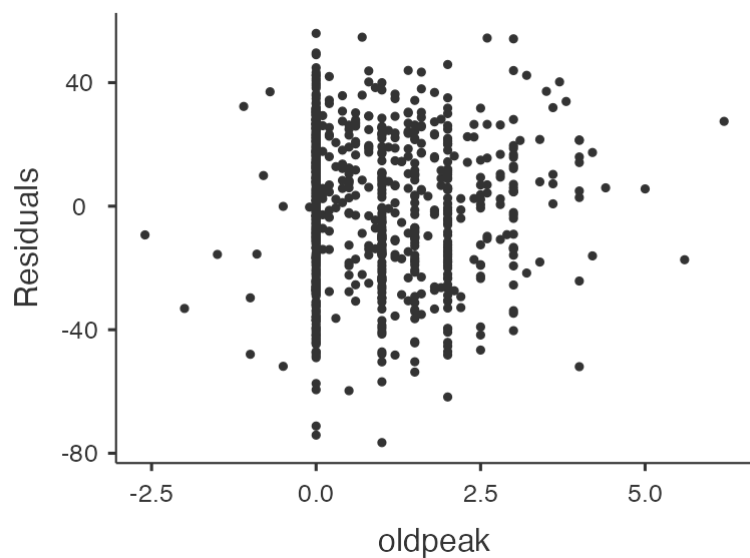
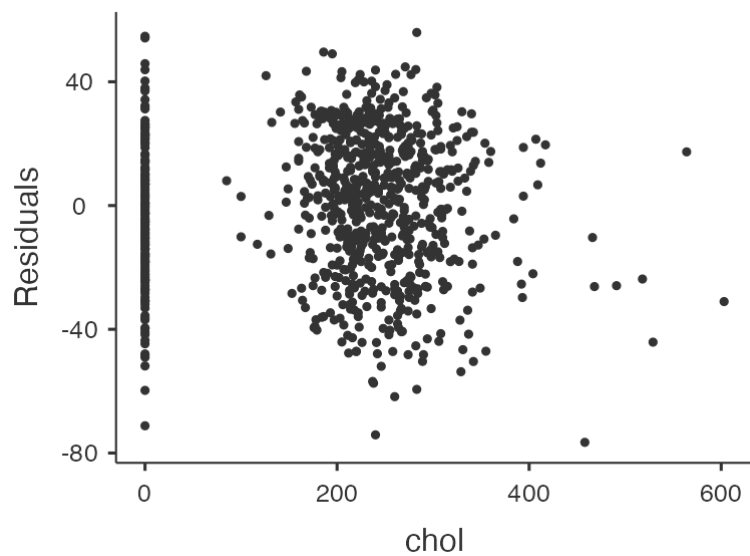
Q-Q Plot



Residuals Plots







ANÁLISIS 9: REGRESIÓN LOGÍSTICA (VARIABLE DEPENDIENTE LIMITADA)

Model Fit Measures

Model	Deviance	AIC	R^2_{McF}
1	726	744	0.369

Note. Models estimated using sample size of N=838

Predictor	Estimate	95% Confidence Interval		SE	Z	p	Odds ratio
		Lower	Upper				
Constante	-0.06872	-1.96014	1.82271	0.96503	-0.0712	0.943	0.934
age	0.03817	0.01701	0.05932	0.01079	3.5362	<.001	1.039
chol	-0.00356	-0.00536	-0.00177	9.15e-4	-3.8901	<.001	0.996
thalch	-0.01164	-0.01975	-0.00352	0.00414	-2.8105	0.005	0.988
sex:							
Male – Female	1.20018	0.74611	1.65425	0.23167	5.1805	<.001	3.321
cp:							
atypical angina – asymptomatic	-2.20635	-2.75816	-1.65454	0.28154	-7.8367	<.001	0.110
non-anginal – asymptomatic	-1.59238	-2.04473	-1.14004	0.23079	-6.8996	<.001	0.203
typical angina – asymptomatic	-0.80375	-1.52865	-0.07886	0.36985	-2.1732	0.030	0.448
exang:							
TRUE – FALSE	1.34168	0.92842	1.75494	0.21085	6.3631	<.001	3.825

Note. Estimates represent the log odds of "enfermedad_binaria = 1" vs. "enfermedad_binaria = 0"

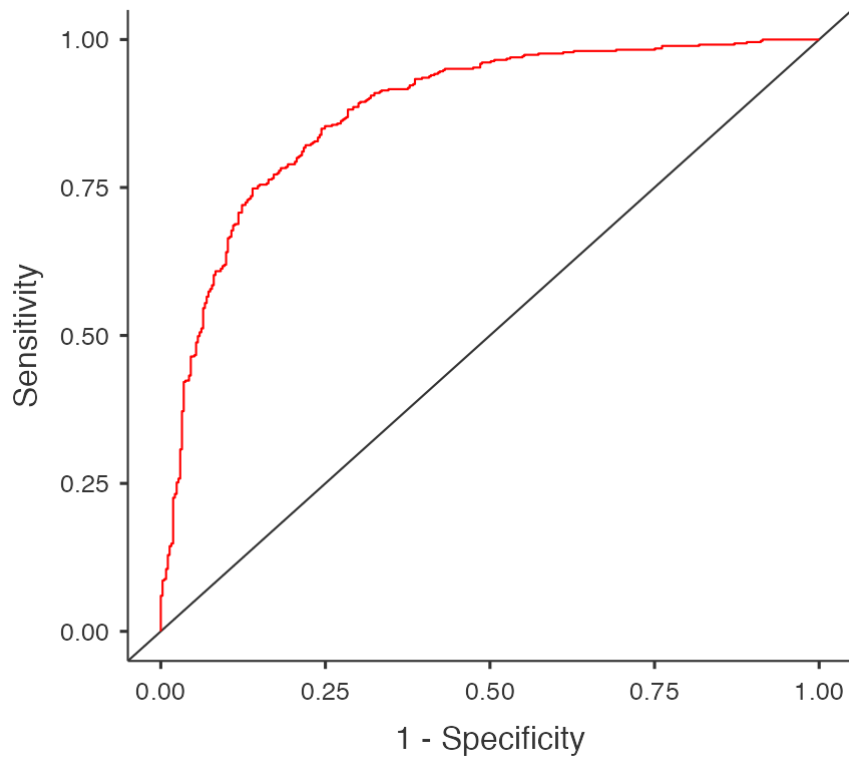
Prediction

Classification Table – ...

Observed	Predicted		% Correct
	0	1	
0	282	91	75.6
1	75	390	83.9

Note. The cut-off value is set to 0.5

ROC Curve



[6]

Contingency Tables

Contingency Tables

sex	cp				Total
	asymptomatic	atypical angina	non-anginal	typical angina	
Female	70	61	53	10	194
Male	426	113	151	36	726
Total	496	174	204	46	920

χ^2 Tests

	Value	df	p
χ^2	37.9	3	<.001
N	920		

References

- [1] The jamovi project (2024). *jamovi*. (Version 2.6) [Computer Software]. Retrieved from <https://www.jamovi.org>.
- [2] R Core Team (2024). *R: A Language and environment for statistical computing*. (Version 4.4) [Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2024-08-07).
- [3] Revelle, W. (2023). *psych: Procedures for Psychological, Psychometric, and Personality Research*. [R package]. Retrieved from <https://cran.r-project.org/package=psych>.
- [4] Revelle, W. (2025). *psych: Procedures for Psychological, Psychometric, and Personality Research*. [R package]. Retrieved from <https://cran.r-project.org/package=psych>.

[5] Fox, J., & Weisberg, S. (2023). *car: Companion to Applied Regression*. [R package]. Retrieved from <https://cran.r-project.org/package=car>.

[6] Sing, T., Sander, O., Beerenwinkel, N., & Lengauer, T., Unterthiner, T., & Ernst, F. G. M. (2020). *ROCR: Visualizing the Performance of Scoring Classifiers*. [R package]. Retrieved from <https://cran.r-project.org/package=ROCR>.